



DTM-CC016A Single Antenna Multi-Channel Transceiver Gateway Module

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单天线多通道收发网关模组

Overview

DTM-CC016A gateway module is a single-antenna multi-channel transceiver gateway module developed by Shanghai Panqi Microelectronics Co., Ltd. based on the PAN3029 chip. The module has a built-in M4 processor, integrated PA and LNA, and can efficiently process Chirp-IoT data transmission and reception. The module operates in the 470-510MHz frequency band and supports rate adaptation and communication data AES encryption.

Users can drive the module through the SPI interface. The gateway module supports RF data reception, data transmission, parameter configuration and other interfaces, which is convenient for users to use. The gateway module can be widely used in one-to-many star networks, supports large-capacity terminal node networks, supports ChirpLAN Mode A/B/C protocols, supports multiple application scenarios, etc.

Key Features

- Operating frequency band: 470~510MHz
- Modulation: Chirp-IoT
- Channels: 8RX+1TX, half-duplex
- Communication rate: Support 8 Chirp-IoT data channels, adaptive
- Receiving sensitivity: -142dBm@SF12 BW62.5kHz
- Maximum transmit power: 28dBm
- Encryption mode: AES hardware encryption
- Communication interface: SPI
- RSSI detection and output
- ChirpLAN Mode A/B/C protocols

Application Areas

- Smart city
- Smart rubber tapping
- Smart farm
- Smart power and other industrial scenarios



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1 Hardware

1.1 Physical Picture



Figure 1-1 Module physical picture





Figure 1-2 Gateway physical picture (Raspberry Pi + baseboard + module)

□ □ □



1.3 Module dimensions

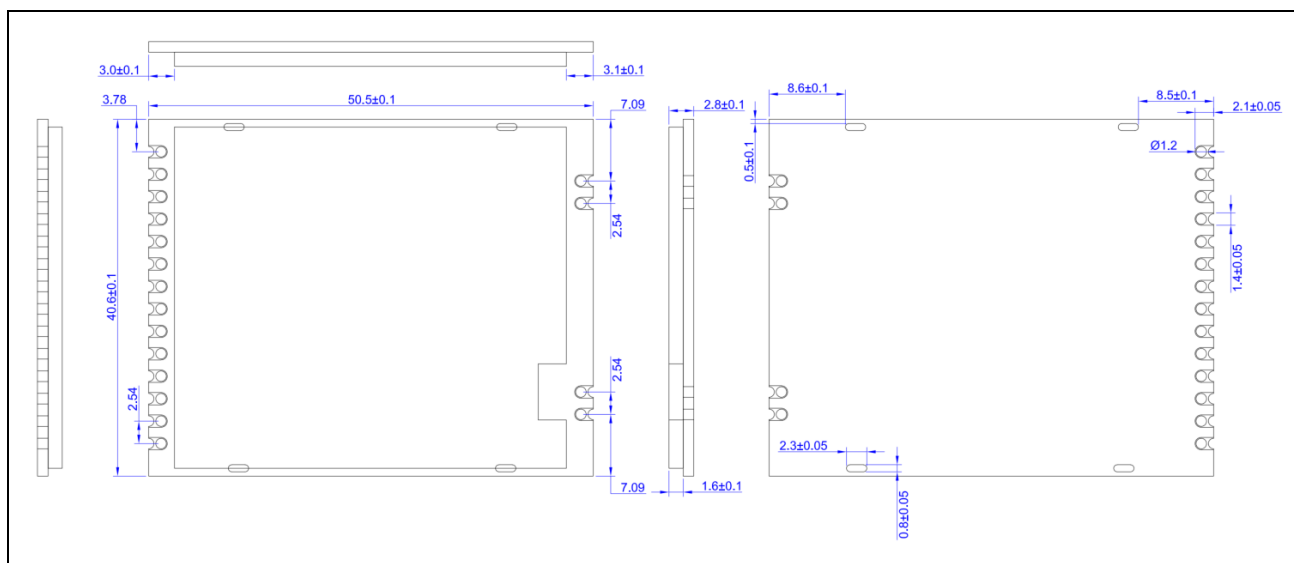


Figure 1-4 Gateway module dimensions

1.4 Module Pin Definition

Port	Description
SPI communication port	1 SPI CSN1_0, MOSI1, MISO1, maximum rate 10MHz
IO1	Custom function pin
Reset	Reset port, RST, 0 active
Programming port	SWD, SWCLK
Serial communication port	UART TX, RX
Power supply	VCC 1PIN, GND 2PIN
Power control	PWR_EN module power switch control pin; 0 - off, 1 - on

1.5 Module baseboard interface definition

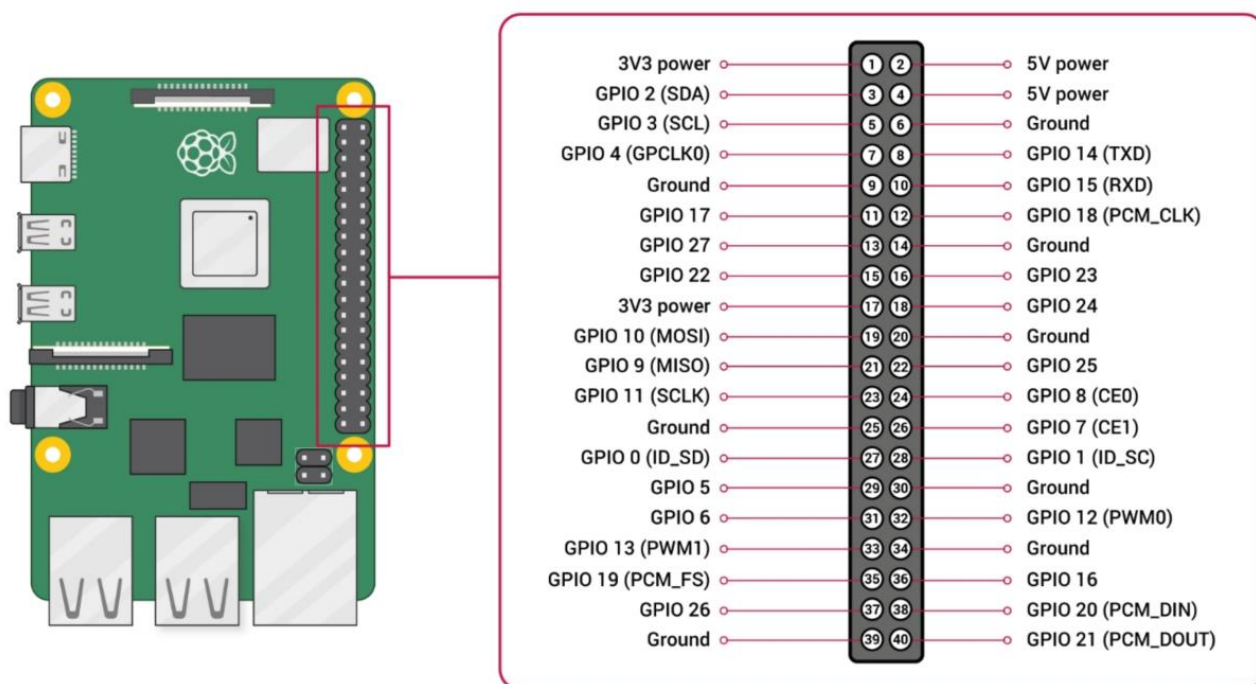


Figure 1-5 Gateway module baseboard pin diagram

1.6 Electrical parameters

Parameter	Description
Power supply voltage	Nominal voltage 5V/2A, input range (4.5V-18V)
Receive current	72mA
Transmit current	870mA@28dBm,470MHz
RF interface	50Ω

1.7 Physical properties

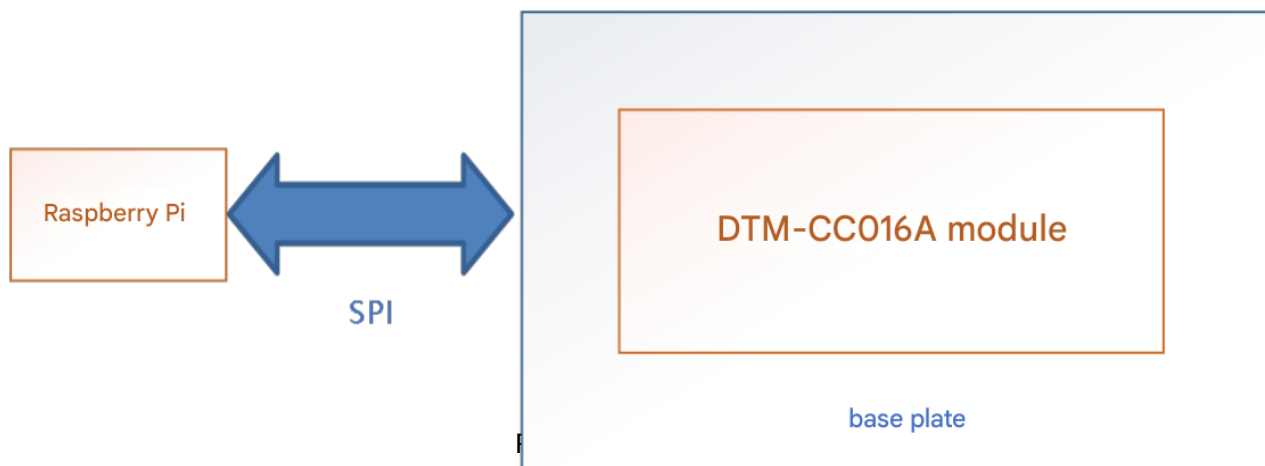
Parameter	Description
Size	50.5*40.6mm
Package	Stamp hole
Temperature	-40~85°C (operating), -55~105°C (storage)
Humidity	5~95% (no condensation)

1.8 RF Parameters

Parameter	Description
Output power	Maximum 28dBm
Bandwidth	62.5kHz, 125kHz, 250kHz, 500kHz
Spreading factor	SF5~SF12
Receiver sensitivity	-142dBm@SF12 BW62.5kHz
Frequency range	470~510MHz
Channels	8 ChirpLoT receive channels, 8RX+1TX half-duplex

2 Module Protocol

The gateway module provides an SPI interface to the outside world. The module acts as an SPI slave. Our company provides Raspberry Pi master control operation examples and gateway module programs. The open source address is: <https://gitee.com/chirplan>



The SPI communication protocol between the module and the main control MCU (Raspberry Pi) is as follows:

(1) The Raspberry Pi obtains the RF data received by the module. The received data content format is as follows:

Receive data area			
Area description	Data segment starting address	Data storage (buffer size)	illustrate
Receive data FIFO check area	0	buf[0] receiving area data size lower 8 bits	The SPI master reads, and when the data length that it is equal to 0, it knows the data in the receiving area, and then writes the data in each upper and lower capacity to the area.
	1	buf[1] receiving area data size high 8 bits	
Receive data FIFO data area	2	0xxx	Packet number (in the protocol example, this value is 2)
	3	payload_size	payload length
	0+4	PAYLOAD	PAYLOAD DATA
	...		
	...		
	...		
	payload_size+3	Channel	Channel number (0-7)
	payload_size+4		
	5+payload_size		
	6+payload_size		Spreading factor 5-12, CR:1-4, CRC:0-1 Bandwidth (6/7/8/9 represent 62.5, 125, 250, 500 respectively) LDR0/1
	7+payload_size	RSSI	
		payload_size	signal quality
		PAYLOAD	
		Channel	Channel number (0-7)
		SF[7:4]CR[3:1]CRC_EN[0]	
		BW[7:4]LDR[0]	
		RSSI	
		CRC_L	CRC16 lower 8 bits (all bytes above)
		CRC_H	CRC16 high 8 bits

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(2) The master control needs to send data through the gateway module.
The protocol format is as follows:

Receive data area					
Area description	Data segment starting address	Data storage (buffer size)	illustrate		
Receive data FIFO check area	0	buf[0] receiving area data size lower 8 bits	The DT master reads, and when the slave judges that it is equal to 0, it places the data in the receiving area, and then writes the data in both upper and lower case to this area.		
	1	buf[1] receiving area data size high 8 bits			
Receive data FIFO data area	2	0xxx	Packet number (in the protocol example, this value is 2)	Buffer size	
	3	payload_size	payload length		
	0+4	PAYLOAD	PAYLOAD DATA		
	...				
	...				
	...				
	payload_size+3				
	payload_size+4	Channel	Channel number (0-7)		
	5+payload_size	SF[7:4]CR[3:1]CRC_EN[0]	Spreading factor 5-12, CR:1-4, CRC:0-1		
	6+payload_size	BW[7:4]LDR[0]	Bandwidth (6/7/8/9 represent 62.5, 125, 250, 500 respectively) LDR0/1		
	7+payload_size	RSSI	signal quality		
		payload_size			
		PAYLOAD			
		Channel			
		SF[7:4]CR[3:1]CRC_EN[0]			
		BW[7:4]LDR[0]			
		RSSI			
		CRC_L	CRC16 lower 8 bits (all bytes above)		
	CRC_H	CRC16 high 8 bits			

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(3) The master controller configures parameters for the gateway module using the following protocol format:

Parameter configuration instruction	data storage	description
0	CMD_CONFIG(0x0A)	Parameter Configuration area
1	buffer_size[0]	Length low byte of the entire packet
2	buffer_size[1]	The length of the entire packet high byte
3	p_center_freq	Pre-stage intermediate frequency (reference value 860K) (default value 1000000)
4		
5		
6		
7	filter_bw	Filter bandwidth (0:500K, 1:1M, 2:2M) (default 2)
8	AGC_ENABLE	Is AGC enabled? 1: Enable, 0: Disable (default value 0)
9	AGC_REG1	Fixed gain 1 (default value 0xd6)
10	AGC_REG2	Fixed gain 2 (default value 0xf9)
11	AGC_REG3	Fixed gain 3 (0x3c)
12	SF0[7:4]CR0[3:1]CRC_EN0[0]	Channel 0 channel parameters SF: 5-12, Smart Search (0: SF5-8, 1: 6-9, 2: 7-10, 3: 8-11, 4: 9-12) CR: 1-4, CRC: 0-1 (Default value 0x79)
13	BW0[7:4]LDR0[0]	(Default value 0x70)
14	FREQ0	Channel 0 frequency, unit: Hz, low byte first (default value 484500000)
15		
16		
17		
18	SF1[7:4]CR1[3:1]CRC_EN1[0]	(Default value 0x79)
19	BW1[7:4]LDR1[0]	(Default value 0x70)
20	FREQ1	Channel 1 frequency, unit: Hz, low byte first (default value 484200000)
21		
22		
23		
24	SF2[7:4]CR2[3:1]CRC_EN2[0]	(Default value 0x79)
25	BW2[7:4]LDR2[0]	(Default value 0x70)
26	FREQ2	Channel 2 frequency, unit: Hz, low byte first (default value 483900000)
27		
28		
29		
30	SF3[7:4]CR3[3:1]CRC_EN3[0]	(Default value 0x79)
31	BW3[7:4]LDR3[0]	(Default value 0x70)
32	FREQ3	Channel 3 frequency, unit: Hz, low byte first (default value 483600000)
33		
34		
35		
36	SF4[7:4]CR4[3:1]CRC_EN4[0]	(Default value 0x79)
37	BW4[7:4]LDR4[0]	(Default value 0x70)
38	FREQ4	Channel 4 frequency, unit: Hz, low byte first (default value 483300000)
39		
40		
41		
42	SF5[7:4]CR5[3:1]CRC_EN5[0]	(Default value 0x79)
43	BW5[7:4]LDR5[0]	(Default value 0x70)
44	FREQ5	Channel 5 frequency, unit: Hz, low byte first (default value 483000000)
45		
46		
47		
48	SF6[7:4]CR6[3:1]CRC_EN6[0]	(Default value 0x79)
49	BW6[7:4]LDR6[0]	(Default value 0x70)
50	FREQ6	Channel 6 frequency, unit: Hz, low byte first (default value 482700000)
51		
52		
53		
54	SF7[7:4]CR7[3:1]CRC_EN7[0]	(Default value 0x79)
55	BW7[7:4]LDR7[0]	(Default value 0x70)
56	FREQ7	Channel 7 frequency, unit: Hz, low byte first (default value 482400000)
57		
58		
59		
60	CRC_L	CRC16 lower 8 bits (all bytes above)
61	CRC_H	CRC16 high 8 bits

Buffer size

Revision History

Version	Date	Content
1.0	2023.09	First edition

Documentation

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